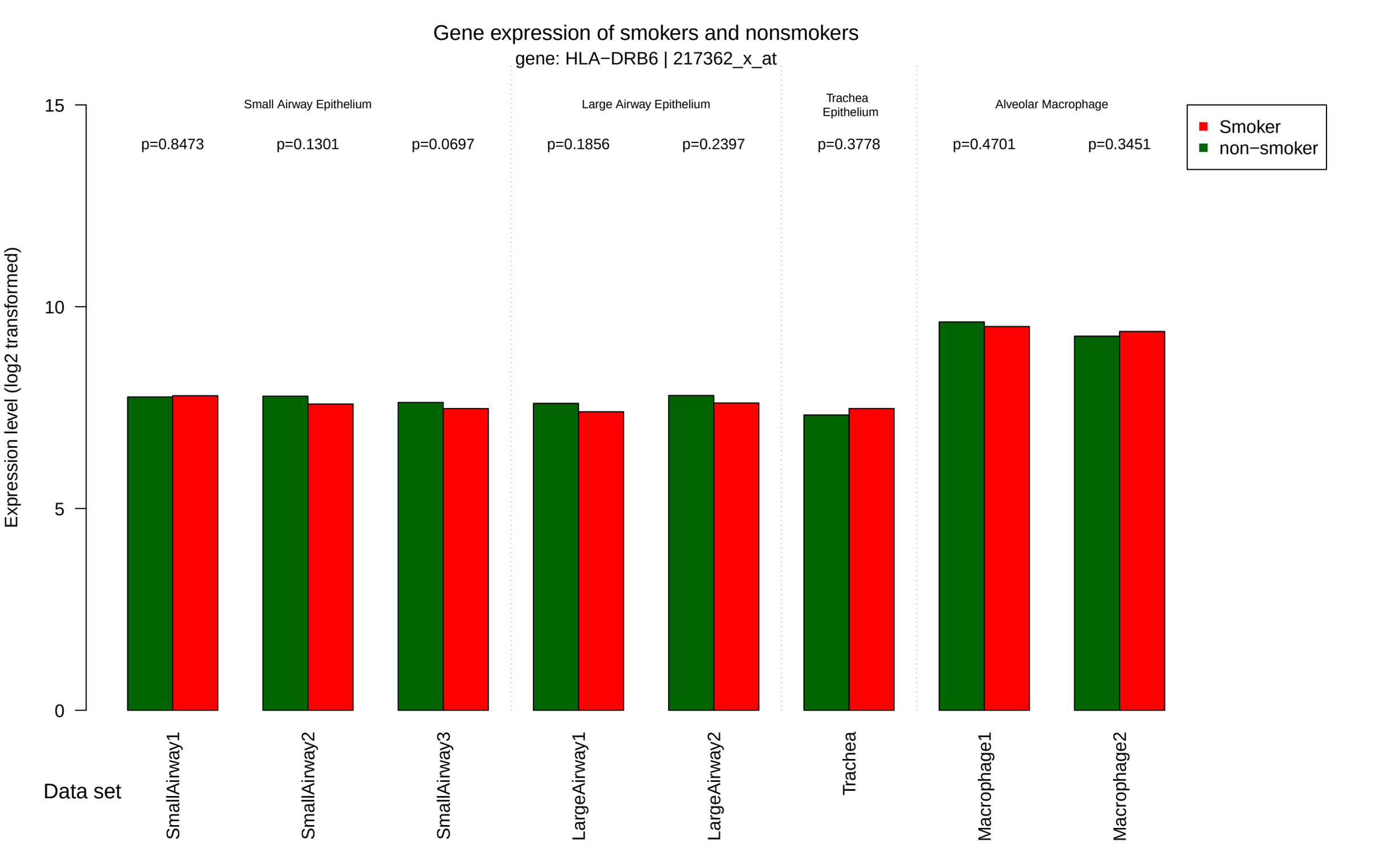
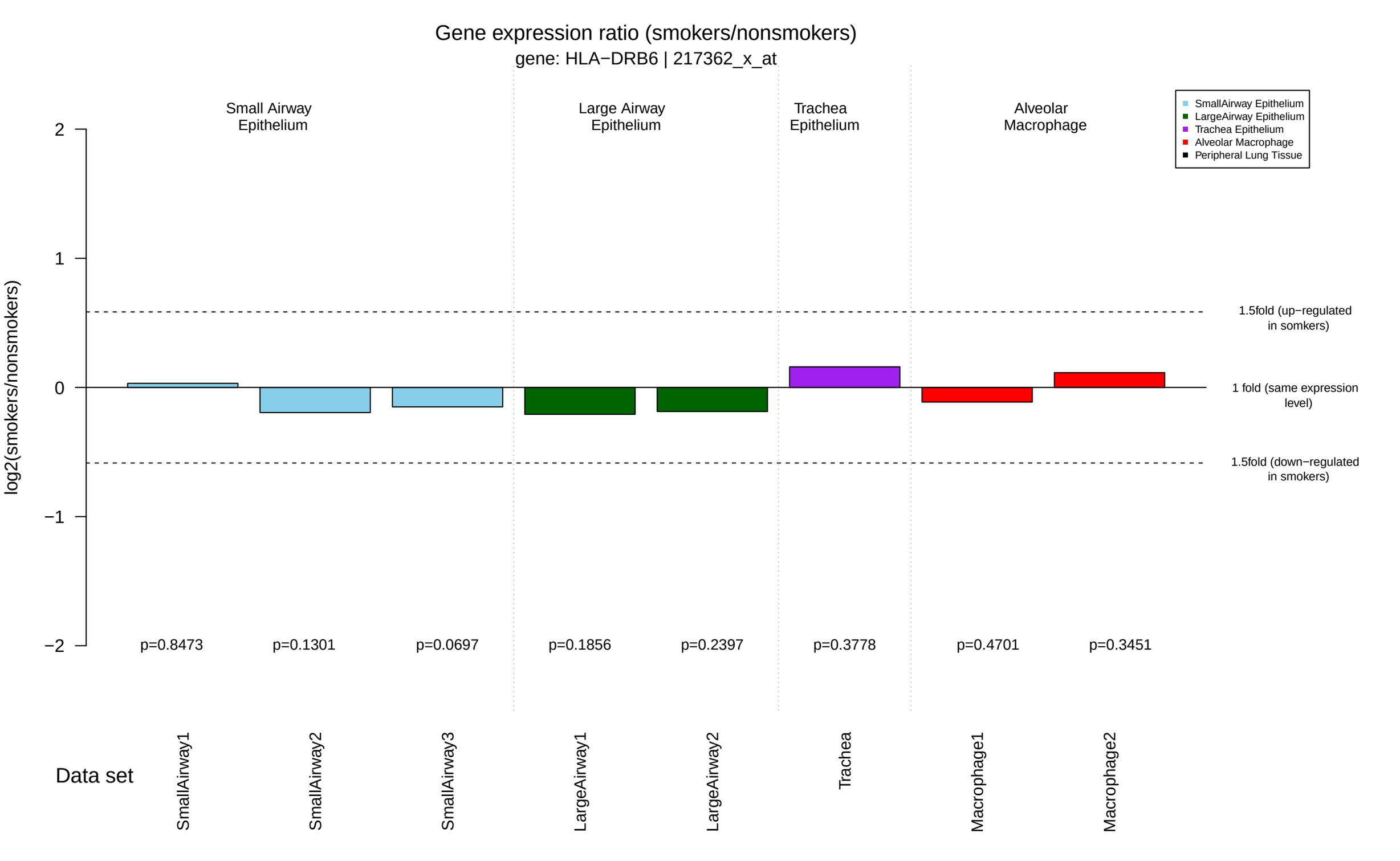
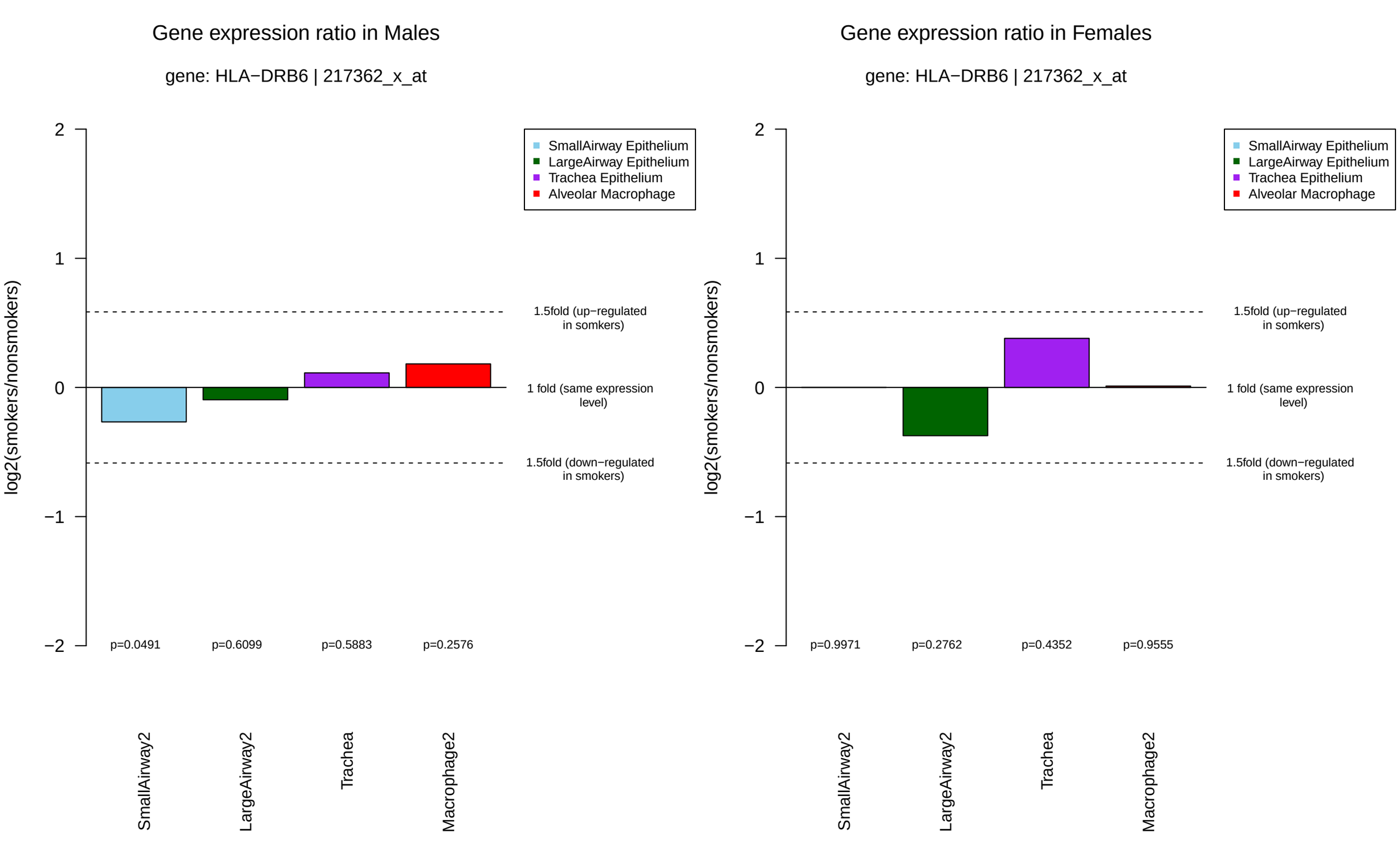
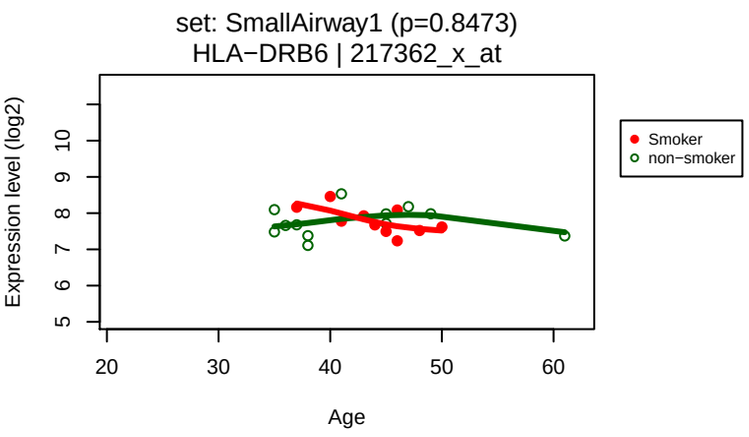
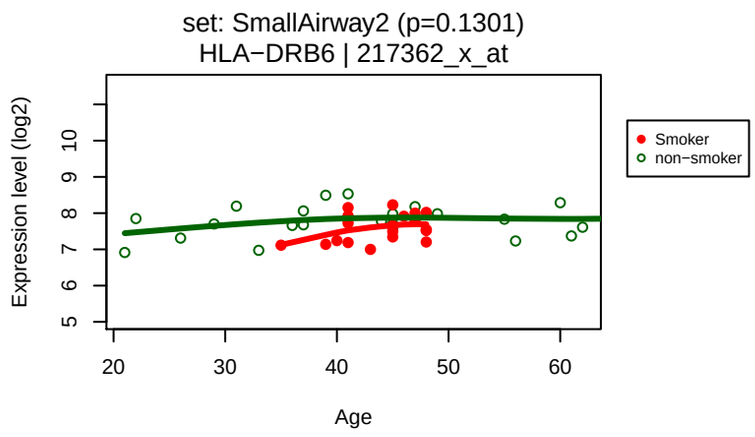
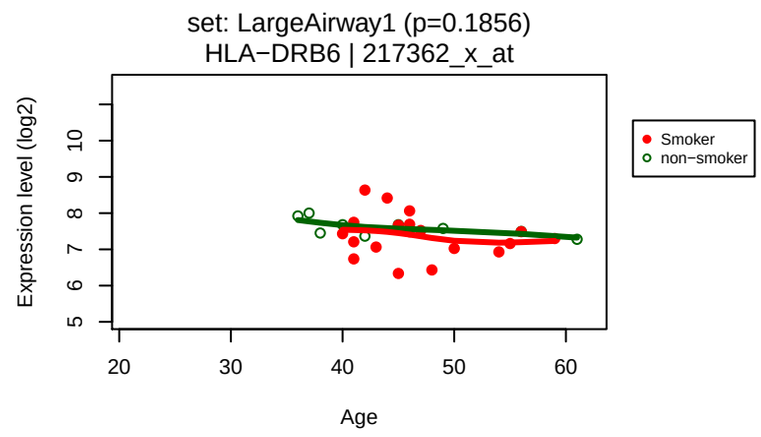

Gene expression of smokers and nonsmokers
gene: HLA−DRB6 | 217362_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8473
p=0.1301
p=0.0697
p=0.1856
p=0.2397
p=0.3778
p=0.4701
p=0.3451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

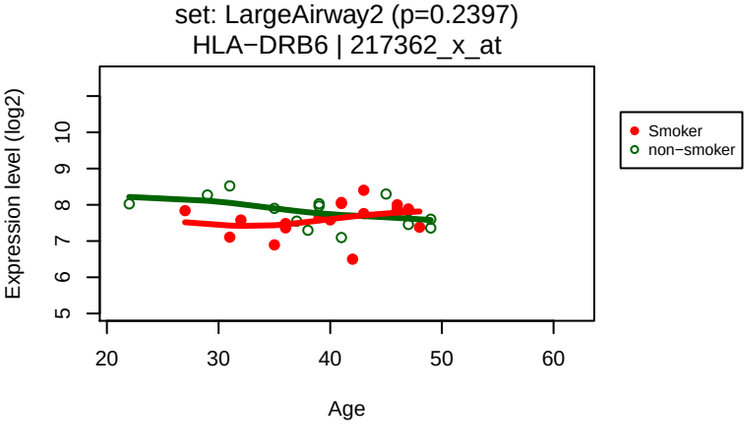

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB6 | 217362_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8473
p=0.1301
p=0.0697
p=0.1856
p=0.2397
p=0.3778
p=0.4701
p=0.3451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

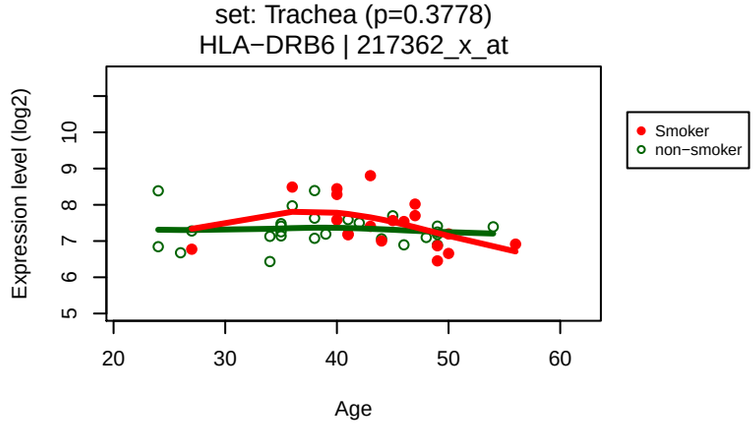

Gene expression ratio in Males
gene: HLA−DRB6 | 217362_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0491
p=0.6099
p=0.5883
p=0.2576
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB6 | 217362_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9971
p=0.2762
p=0.4352
p=0.9555
SmallAirway2
LargeAirway2
Trachea
Macrophage2

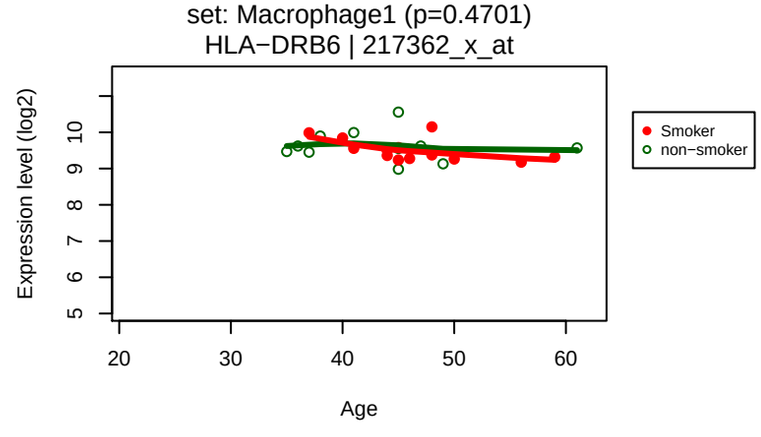

set: SmallAirway1 (p=0.8473)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker

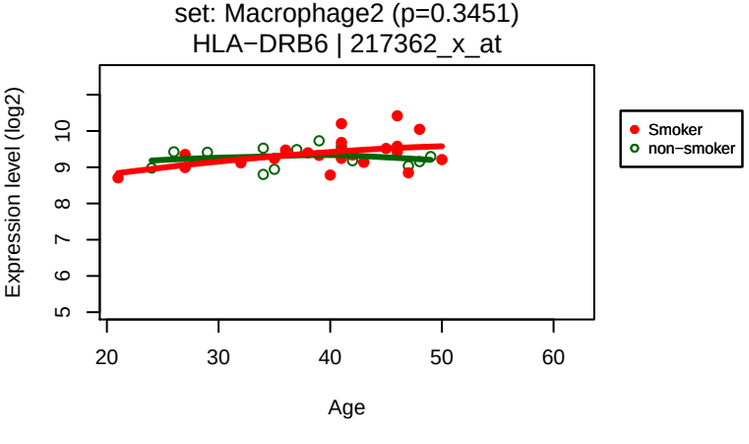

set: SmallAirway2 (p=0.1301)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker

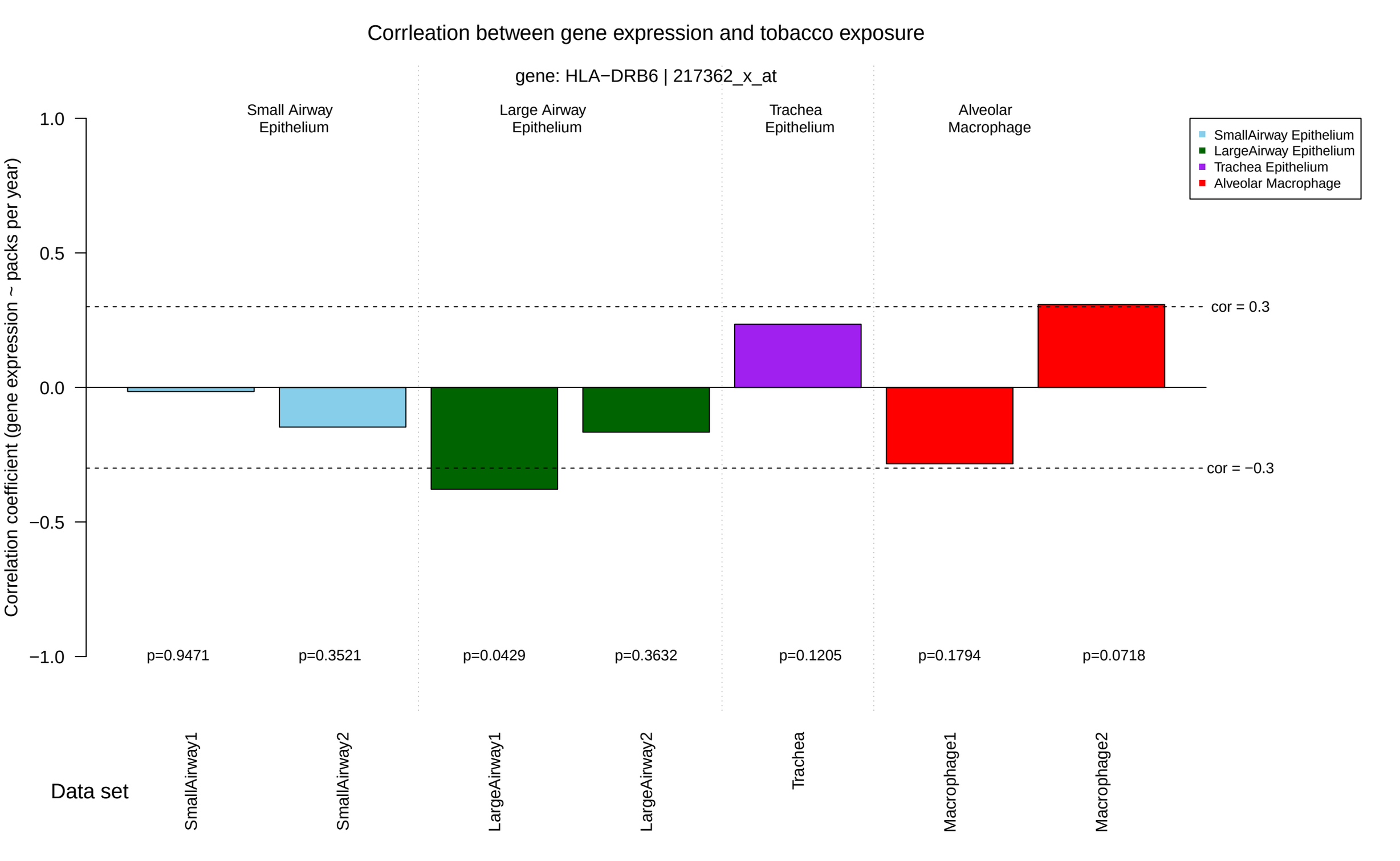

set: LargeAirway1 (p=0.1856)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2397)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3778)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4701)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3451)
HLA−DRB6 | 217362_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB6 | 217362_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9471
p=0.3521
p=0.0429
p=0.3632
p=0.1205
p=0.1794
p=0.0718
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2